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<HOTEL BOOKING SYSTEM>
SOFTWARE DESIGN DESCRIPTION DOCUMENT

VERSION 1.0

Sharjeel Abid

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Information Technology

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Document Approval

The following Software Requirements Specification has been accepted and approved by the following:

Signature	Printed Name	Title	Date
	Mam Kainat		2026-01-10

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1. INTRODUCTION

1.1 Purpose

This document explains the architecture, design structure, components, data flow, and major algorithms used in the *Hotel Booking System*. It records key technical decisions so that developers, testers, maintainers, supervisors, and other stakeholders can understand, implement, and maintain the system. It also converts the SRS requirements into an organized technical design for development and testing.

Intended Audience:

- Developers (frontend, backend, database)
- QA/testing team
- Project supervisor / examiners
- System maintainers and administrators
- Technical stakeholders (e.g., product owner)

1.2 Scope

Description:

The Hotel Booking System is a web-based platform that allows users to browse hotels, check room availability, make or cancel bookings, and pay online. Administrators manage hotels, rooms, prices, bookings, and basic reports. The system also supports login, registration, and confirmation emails.

Main Goals:

- Provide an easy interface for users to search and book rooms
- Maintain accurate room availability
- Allow admins to manage hotels, rooms, and bookings
- Securely handle user and payment data
- Generate simple reports (daily bookings, occupancy)

Main Benefits:

- Faster and automated booking process

- Better customer experience
- Centralized booking data
- Easier hotel management

In Scope:

- User login/registration
- Searching hotels
- Viewing room details
- Booking rooms
- Admin management (CRUD)
- Reports
- Email notifications

1.3 Overview

This document is structured to provide a clear and complete technical design. It includes:

- Introduction (purpose, scope, references, definitions)
- System architecture diagrams
- Component design (frontend, backend, database, services)
- Data design (ER diagram, tables)
- Interfaces (UI and API)
- Sequence diagrams (booking flow, login process)
- Non-functional design (security, performance, scalability)
- Design decisions and trade-offs
- Testing considerations

Each section provides diagrams and descriptions required for development and testing.

1.4 Reference Material

The following resources were used in preparing this design document:

- Hotel Booking System – Software Requirements Specification (v1.0)
- Software engineering lecture notes
- IEEE guidelines for design documents
- Draft ER diagrams and schema
- UI mockups (Figma/Sketch)
- REST API and JSON formatting guidelines
- Security best practices (HTTPS, hashing, OWASP)
- Framework documentation (Django/Flask/Express)

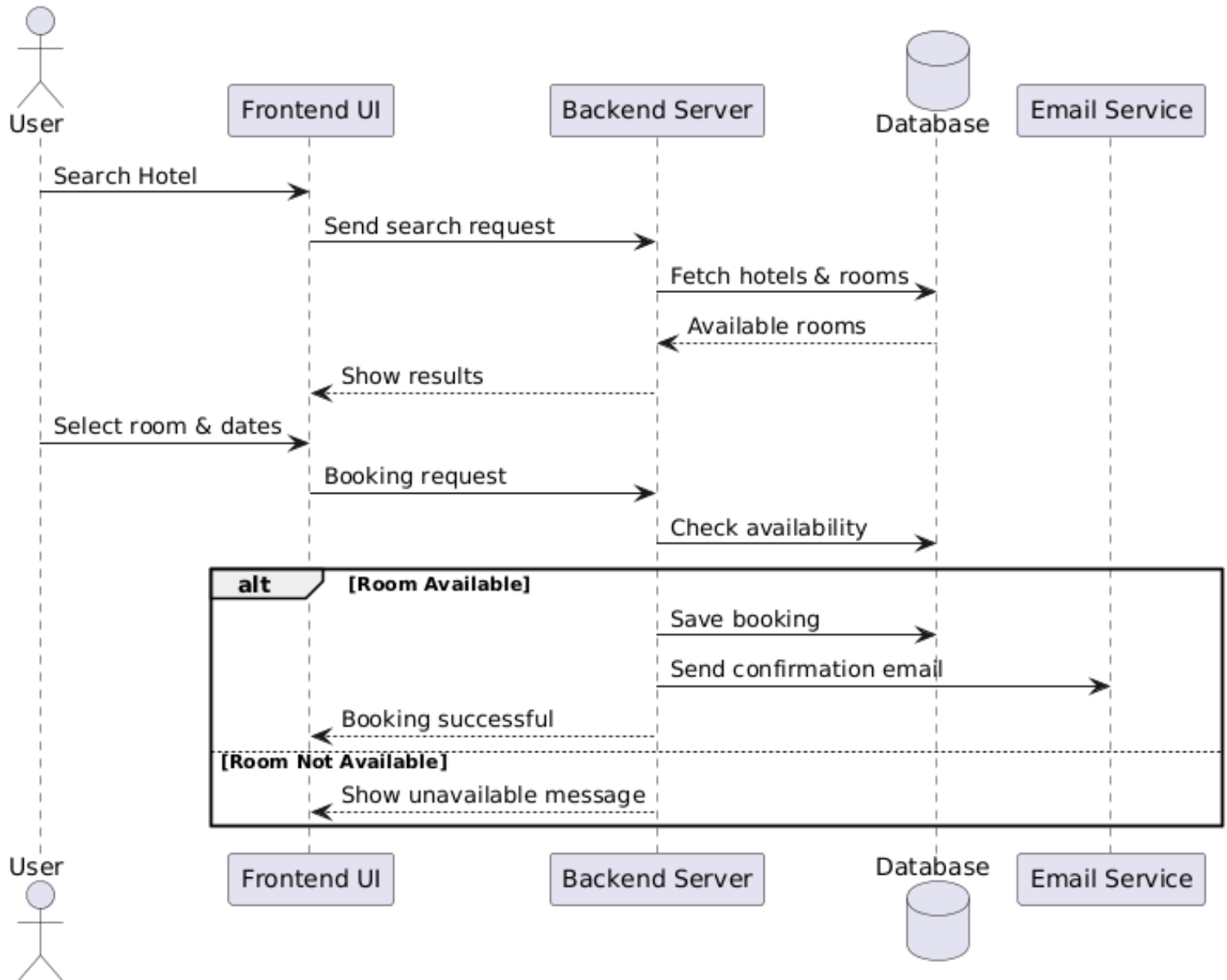
1.5 Definitions and Acronyms

Common Terms:

- **Booking/Reservation:** Confirmed room for selected dates
- **Guest/User:** Customer who books rooms
- **Administrator:** Person managing hotels, rooms, and bookings
- **Room Inventory:** Available rooms and their status
- **Check-in / Check-out:** Arrival and departure dates
- **OTA:** Third-party booking website (Not included)
- **Acronyms:**
- **SRS:** Software Requirements Specification
- **UI/UX:** User Interface / User Experience
- **API:** Application Programming Interface
- **DBMS:** Database Management System
- **CRUD:** Create, Read, Update, Delete
- **REST:** Representational State Transfer
- **JSON:** Data transfer format
- **HTTP/HTTPS:** Web communication protocol
- **MVC:** Architecture pattern
- **ORM:** Database mapping tool

- **SSL:** Secure communication layer
- **OTP:** One-Time Password
- **DB:** Database

Room Booking Sequence Diagram



2. SYSTEM OVERVIEW

The Hotel Booking System is a **web-based application** that allows users to search hotels, check room availability, make or cancel bookings, and receive booking confirmations. Administrators can manage hotels, rooms, prices, bookings, and generate basic reports.

Functionality:

- **Users:** Search hotels, view rooms, book or cancel rooms, receive email confirmations.
- **Admins:** Manage hotels, rooms, prices, bookings, and generate reports.
- **System Design:**
- **Three-tier architecture:**
 - **Frontend:** User interface for customers and admins.
 - **Backend:** Business logic, authentication, and booking management.
 - **Database:** Stores users, hotels, rooms, and bookings.
- **Secure and reliable:** Real-time room availability prevents double bookings; user data is protected.

Background:

Automates hotel bookings, improving efficiency for users and hotel staff while reducing manual errors.

3.1 Architectural Design

The system uses a **three-tier modular architecture** to separate responsibilities:

1. **Frontend (User Interface):** Handles all interactions with users and administrators. Displays hotels, rooms, booking forms, and reports.
2. **Backend (Application Logic):** Processes business rules, manages bookings, authentication, email notifications, and connects frontend with the database.

3. **Database (Data Repository):** Stores information about users, hotels, rooms, bookings, and admin data.

Collaboration:

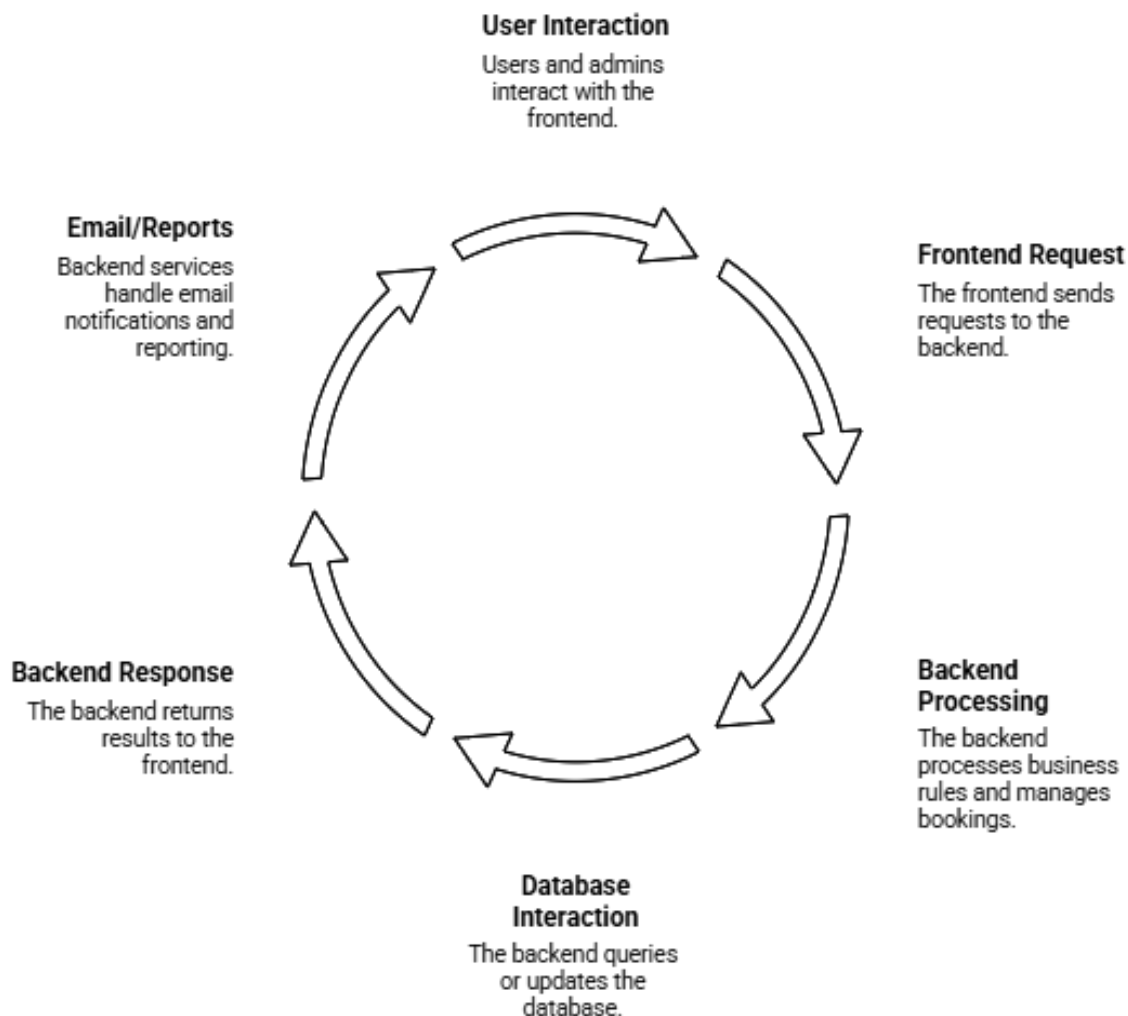
- Users and admins interact with the frontend, which sends requests to the backend.
- The backend queries or updates the database and returns results to the frontend.
- Email notifications and reporting features are handled by backend services.

Diagram (Conceptual Overview):

[User/Admin] → [Frontend (UI)] → [Backend (Logic)] → [Database]

↳ Email/Reports Services

Three-Tier System Interaction Cycle



3.2 Decomposition Description

Subsystems:

1. **User Module:** Handles registration, login, hotel search, booking, and cancellations.
2. **Admin Module:** Handles hotel/room management, booking approval, and reports.
3. **Booking Module:** Manages booking flow, availability checks, and confirmation emails.
4. **Database Module:** Stores all persistent data (users, hotels, rooms, bookings).

Functional Decomposition (DFD - Top Level):

- Input: User/Admin actions
- Process: Authentication, Booking Management, Hotel Management
- Output: Booking confirmation, reports, updated data

3.3 Design Rationale

- **Why three-tier architecture:**
 - Clear separation of concerns (UI, logic, data)
 - Easier maintenance and future upgrades
 - Supports scalability and secure data handling
- **Considered alternatives:**
 - **Monolithic design:** Rejected because changes in one part would affect the whole system.
 - **Micro services:** Overkill for this small project; added unnecessary complexity.

4. DATA DESIGN

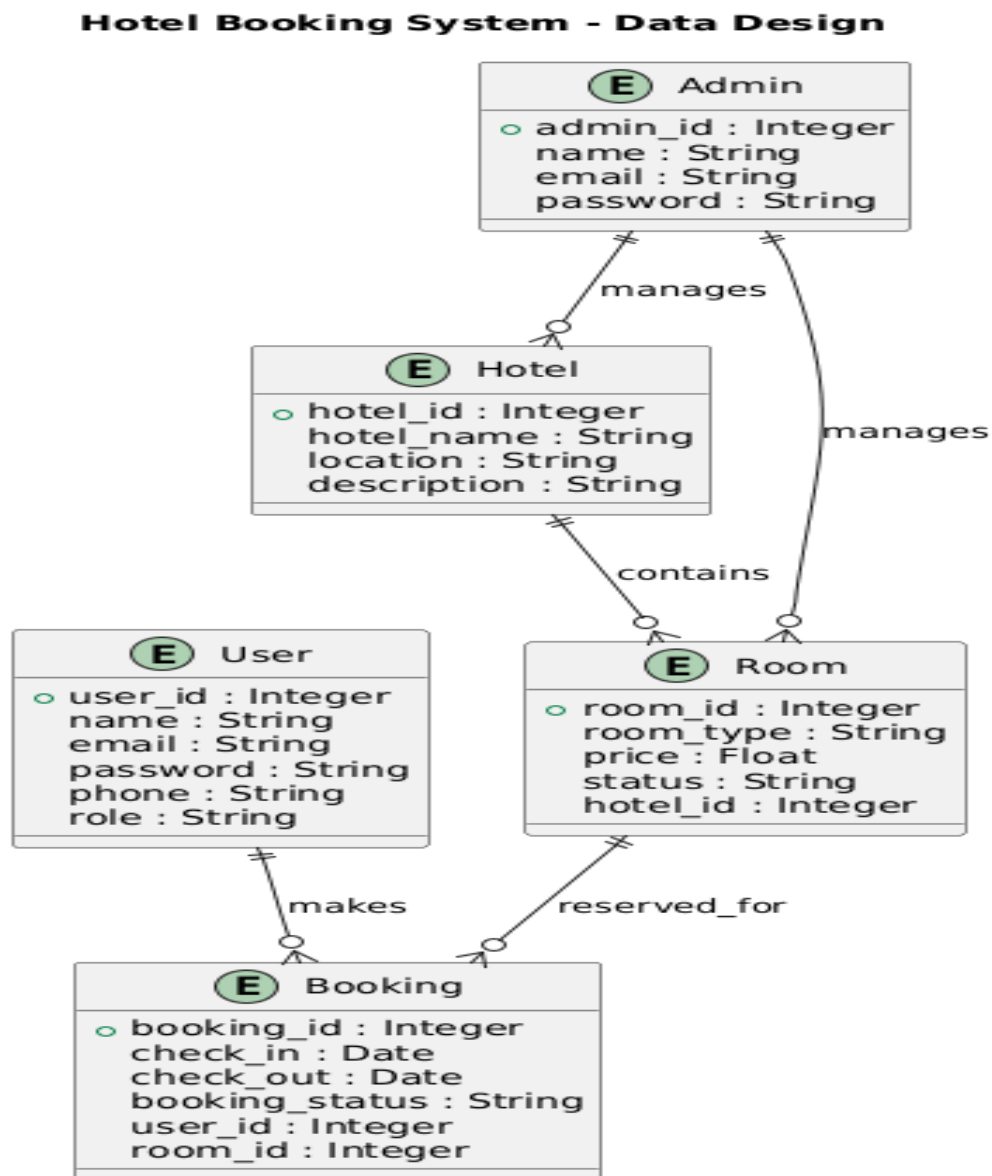
4.1 Data Description

Explain how the information domain of your system is transformed into data structures.

Describe how the major data or system entities are stored, processed and organized. List any databases or data storage items.

4.2 Data Dictionary

Alphabetically list the system entities or major data along with their types and descriptions. If you provided a functional description in Section 3.2, list all the functions and function parameters. If you provided an OO description, list the objects and its attributes, methods and method parameters.



5. COMPONENT DESIGN

5.1 User Module

Responsibility: Handles all user interactions.

Functions:

- **Registration/Login:** Allows users to create accounts and securely log in.
- **Search Hotels:** Users can search hotels by location, dates, and filters.
- **View Rooms:** Displays room details, images, and amenities.
- **Booking & Cancellation:** Users can reserve or cancel rooms.
- **Notifications:** Sends booking confirmations via email.

Local Data: User profiles, search criteria, and current booking data.

5.2 Admin Module

Responsibility: Manages hotel operations and system content.

Functions:

- **Hotel & Room Management:** Add, edit, or remove hotel and room details.
- **Price & Availability Management:** Update room pricing and availability.
- **Booking Management:** View, approve, or cancel bookings.
- **Reports:** Generate basic reports such as daily bookings and occupancy rates.

Local Data: Hotel records, room inventory, booking history, and admin credentials.

5.3 Booking Module

Responsibility: Ensures proper booking flow and availability checks.

Functions:

- **Availability Check:** Confirms rooms are available for selected dates.
- **Booking Confirmation:** Reserves the room and updates the database.
- **Conflict Handling:** Prevents double bookings by checking real-time availability.
- **Notifications:** Triggers emails for successful bookings or cancellations.

Local Data: Booking details, check-in/check-out dates, room status.

5.4 Database Module

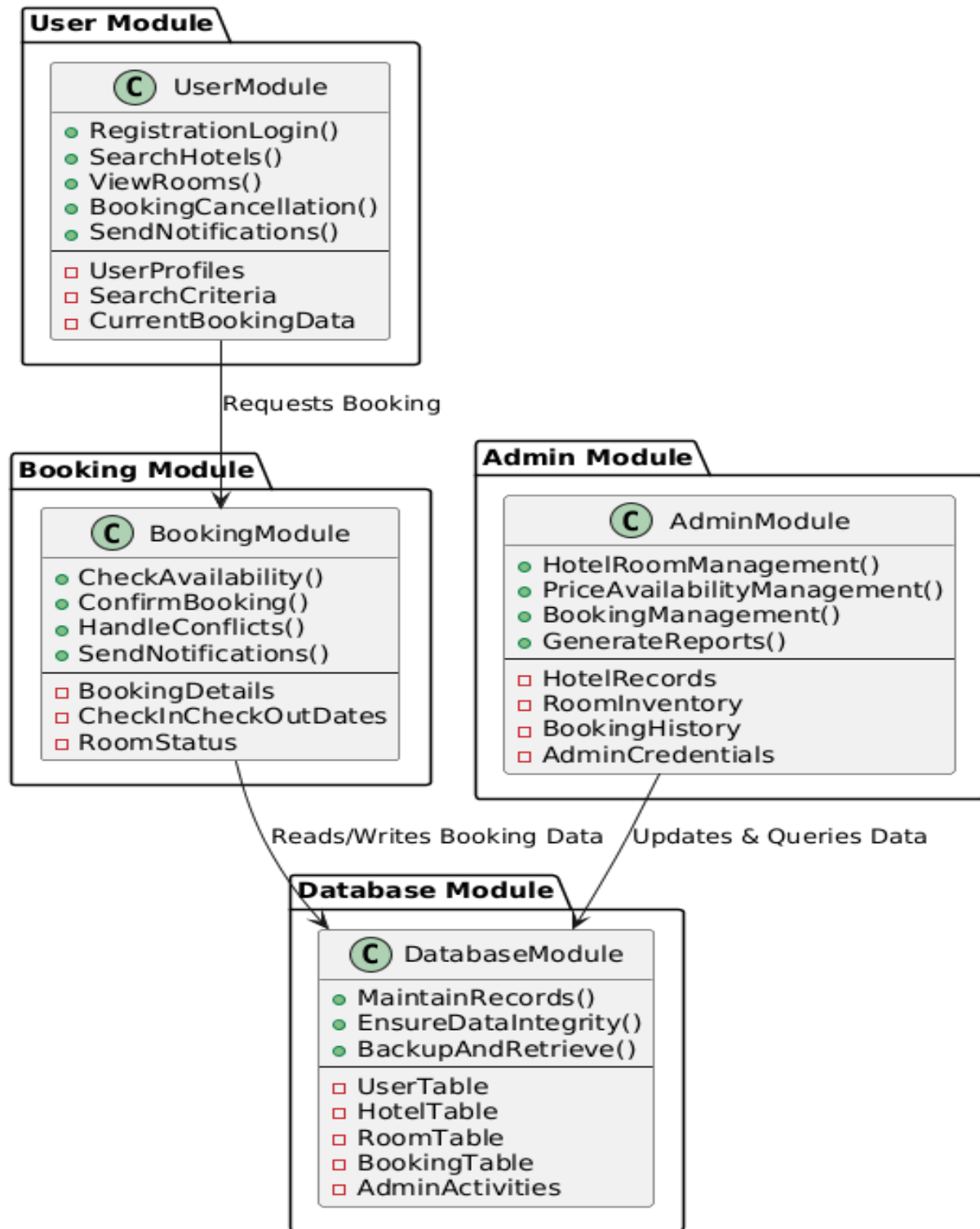
Responsibility: Stores and manages all persistent data.

Functions:

- Maintain records for users, hotels, rooms, bookings, and admins.
- Ensure data integrity and support queries from the backend.
- Provide backup and retrieval for reporting and analytics.

Local Data: Tables for users, hotels, rooms, bookings, and admin activities.

Hotel Booking System - Component Design



6.1 Overview of User Interface

The user interface (UI) is designed to be **simple, intuitive, and responsive** for both users and administrators. Users can easily navigate the system to search hotels, view room details, make or cancel bookings, and receive confirmation messages. Administrators can access management screens to add or update hotels and rooms, manage bookings, and view reports.

Key Features from User's Perspective:

- **Search & Filter:** Users enter location, dates, and preferences to find suitable hotels.
- **Booking Flow:** Users select a room, confirm availability, and reserve it.
- **Feedback:** Confirmation messages, alerts for unavailable rooms, and booking notifications via email.
- **Account Management:** Users can register, log in, and view booking history.

6.2 Screen Images

(Screenshots can be hand-drawn or created using tools; main screens include:)

1. **Home Page:** Search bar, hotel listings, navigation menu.
2. **Hotel Details Page:** Room images, amenities, price, and booking button.
3. **Booking Page:** Select check-in/check-out dates, confirm booking.
4. **Admin Dashboard:** Manage hotels, rooms, bookings, and generate reports.

(Include hand-drawn or digital screenshots in the final report.)

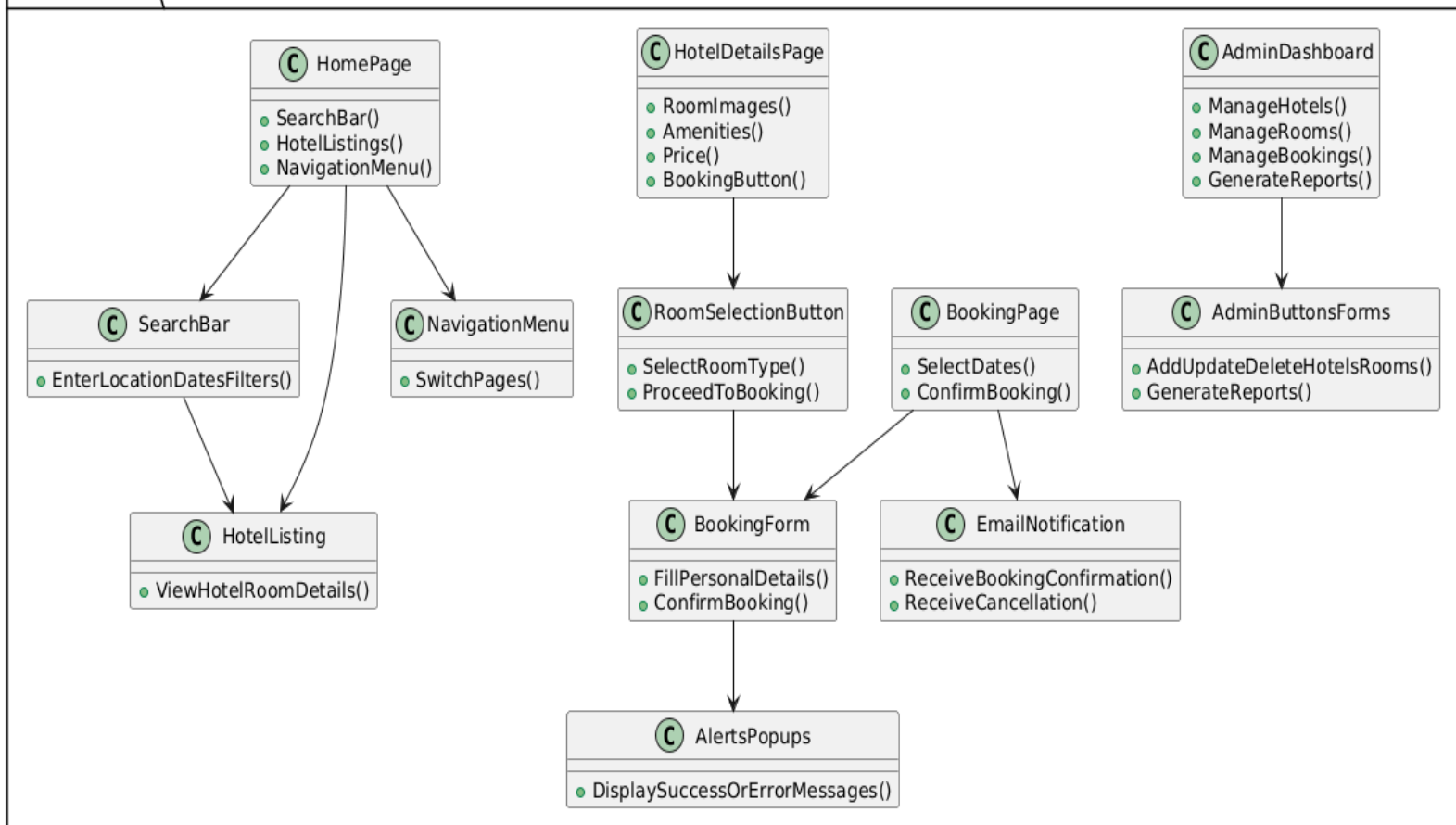
6.3 Screen Objects and Actions

Screen Object	Action / Function
Search Bar	Enter location, dates, and filters to find hotels
Hotel Listing	Click to view details of each hotel and rooms
Room Selection Button	Choose room type and proceed to booking
Booking Form	Fill personal details, confirm booking

Email Notification	Receive confirmation or cancellation details
Admin Buttons/Forms	Add, update, or delete hotels/rooms, generate reports
Navigation Menu	Switch between user pages or admin dashboard
Alerts/Pop-ups	Display errors (e.g., room unavailable) or success messages

Hotel Booking System - User Interface

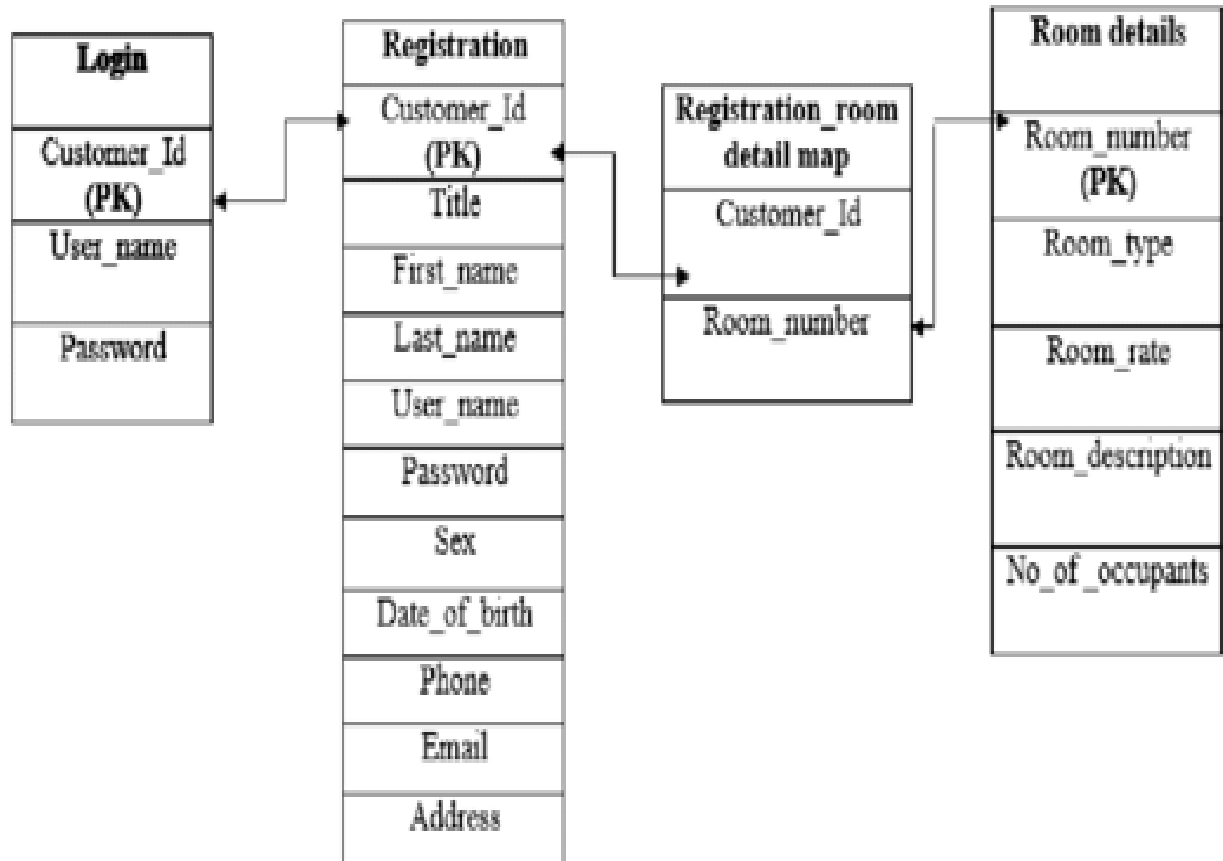
User Interface



7. REQUIREMENTS MATRIX

This table shows how each system component satisfies the functional requirements from the SRS.

SRS Requirement ID	Requirement Description	System Component
FR-1	User can register and log in	User Module
FR-2	User can search hotels by location, date, and filters	User Module / Booking Module
FR-3	User can view hotel and room details	User Module
FR-4	User can make a booking	Booking Module / Database Module
FR-5	User can cancel a booking	Booking Module / Database Module
FR-6	System sends booking confirmation emails	Booking Module / Backend Services
FR-7	Admin can add, edit, or delete hotels	Admin Module / Database Module
FR-8	Admin can manage rooms and pricing	Admin Module / Database Module
FR-9	Admin can view and manage bookings	Admin Module / Database Module
FR-10	Admin can generate basic reports	Admin Module / Backend Services



8. APPENDICES

This section provides **additional supporting details** that help understand the system design. It is optional but useful for reference. Examples of what can be included:

- **Glossary of Terms:** Definitions of technical terms and acronyms used in the document.
- **Revision History:** Track changes made to the design document with dates and version numbers.
- **Deployment Instructions:** Step-by-step instructions for installing or deploying the system.
- **References to Tools and Resources:** Links or file names for UI mockups, ER diagrams, and frameworks used.
- **Sample Data:** Example records for users, hotels, rooms, and bookings to illustrate system usage.

Purpose:

The appendices provide clarity, make the document easier to understand, and act as a reference for developers, testers, and maintainers.